



# The tap was right. It landed too early.

Why a test that does everything right still flakes on mobile, and how Mobium asks the app when it is done. Measured on an emulator, a simulator, a Pixel 8 Pro and an iPhone 15 Plus.

```
row B, generation 1: stale    launched normally
```

```
row B, generation 2: current  with --gray-box
```

# Every wait passed. The tap still missed.

```
$ mobium tap testid=busyQuiet  
$ mobium tap testid=busyRowB  
$ mobium text testid=busyOutcome  
row B, generation 1: stale
```

Before it taps, Mobium waits for the target to pass four checks. Row B passed all four the whole time:

on screen PASSED

still PASSED

enabled PASSED

uncovered PASSED

It was also about to be replaced, and nothing on screen said so.

# Only the app knows it is still busy.

MobiumApp's Busy Demo starts 0.4 to 1.6 seconds of work and leaves the old rows up while it runs. A tap that comes before the new rows lands on a row that is about to go.



On a real Pixel and iPhone, 7 to 9 taps in 10 landed stale.

# So the app says when it is busy.

```
const refresh = () => {  
  GrayBox.busy('quiet');  
  // the work, then the new rows  
};  
// once the new rows have rendered:  
GrayBox.idle('quiet');
```

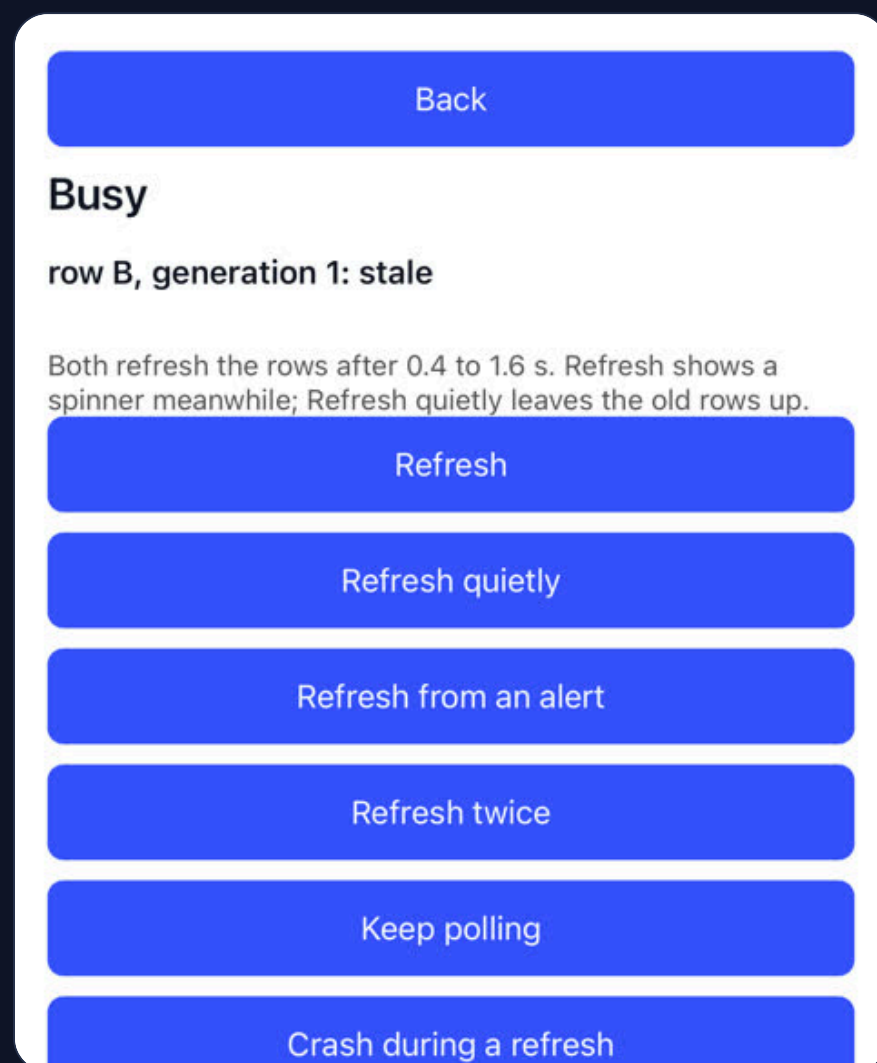
MobiumApp links Mobium's gray-box library, in Swift and Kotlin. It writes one line to the device log for each change, and Mobium already reads that log:

```
MOBIUM-GRAYBOX busy=1 tag=quiet  
MOBIUM-GRAYBOX still busy=1  
MOBIUM-GRAYBOX busy=0 tag=quiet
```

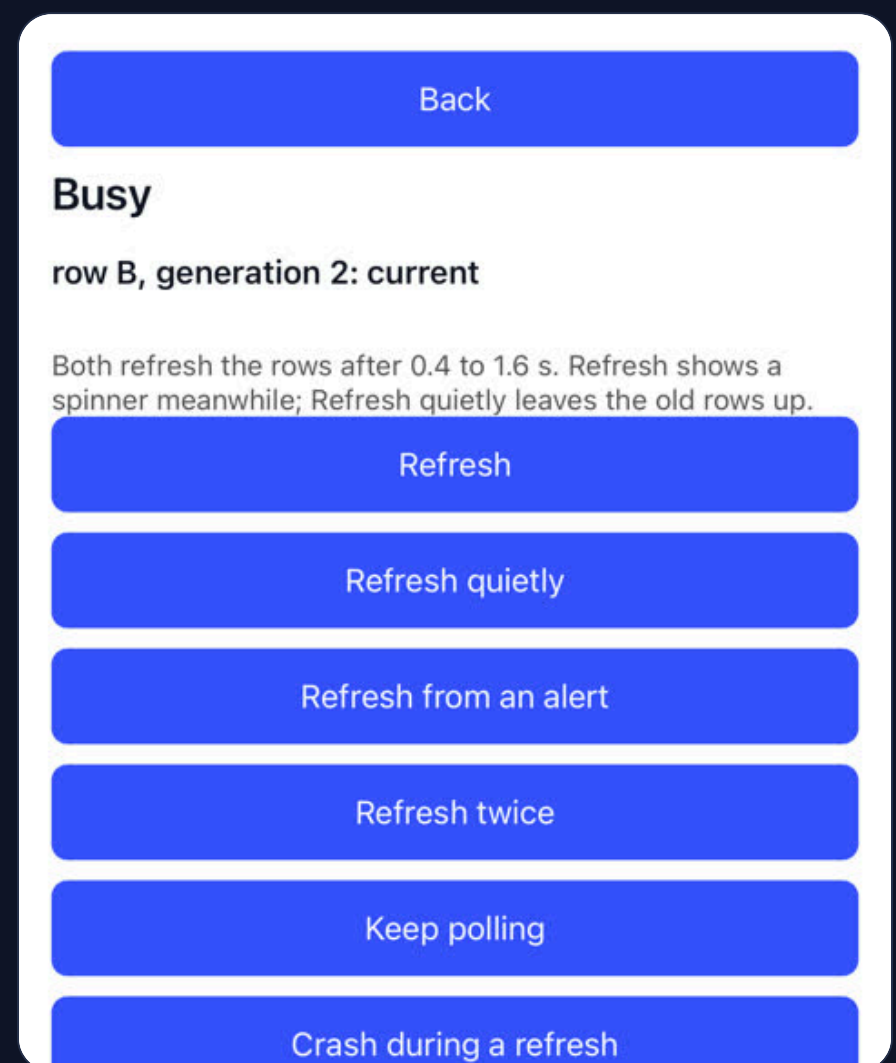
# One flag. The tap waits for the app.

```
$ mobium launch --gray-box dev.mobium.mobiumapp  
$ mobium tap testid=busyRowB  
gray box: waited 955 ms for the app to go idle  
(busy: quiet)
```

## LAUNCHED NORMALLY



## WITH --GRAY-BOX



# The same test, one key apart.

```
"app": "dev.mobium.mobiumapp",  
+ "grayBox": true,
```

Five runs each way, on four devices:

| DEVICE                  | WITHOUT | WITH   | WAITED  |
|-------------------------|---------|--------|---------|
| Pixel 7 emulator        | 0 of 5  | 5 of 5 | 1148 ms |
| iPhone 17 Pro simulator | 0 of 5  | 5 of 5 | 1202 ms |
| Pixel 8 Pro             | 3 of 5  | 5 of 5 | 709 ms  |
| iPhone 15 Plus          | 0 of 5  | 5 of 5 | 955 ms  |

Passing 3 of 5 is what flaky looks like: it depends on how long the work happened to take.

# A wait that knows when to stop.

## Work that never ends

Refused after 10 seconds, naming what kept the app busy.

## The app goes home, or crashes

Not waited on. Busy is a lease the app renews twice a second, and a dead app stops renewing it.

## The log stream drops

The result says Mobium could not hear the app, and it listens again.

## Every result says what it waited for

“waited 955 ms (busy: quiet)”, “the app was idle”, or why it did not wait.

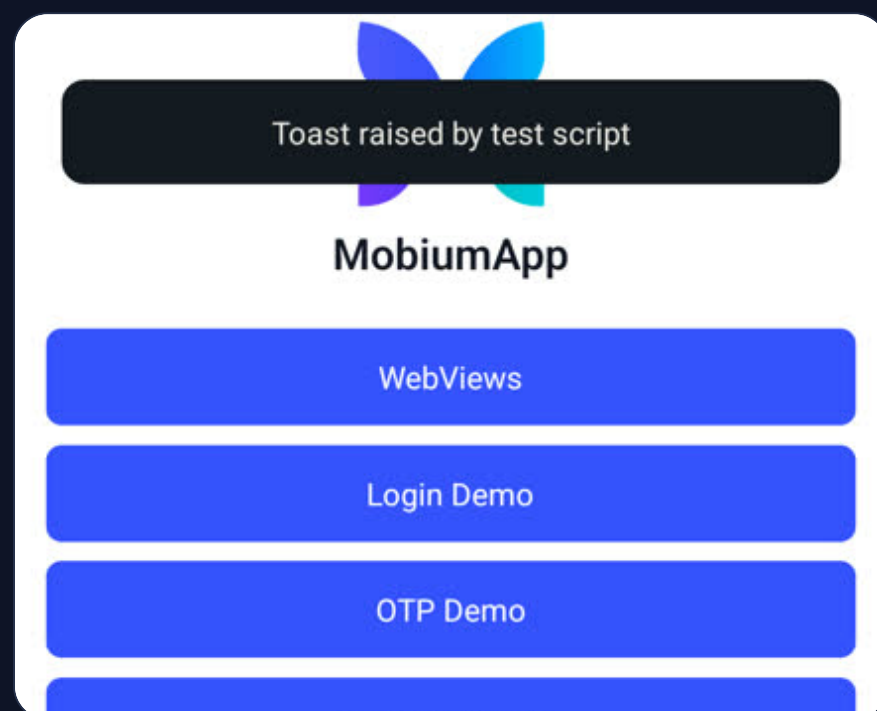
All six edge checks pass on all four devices.



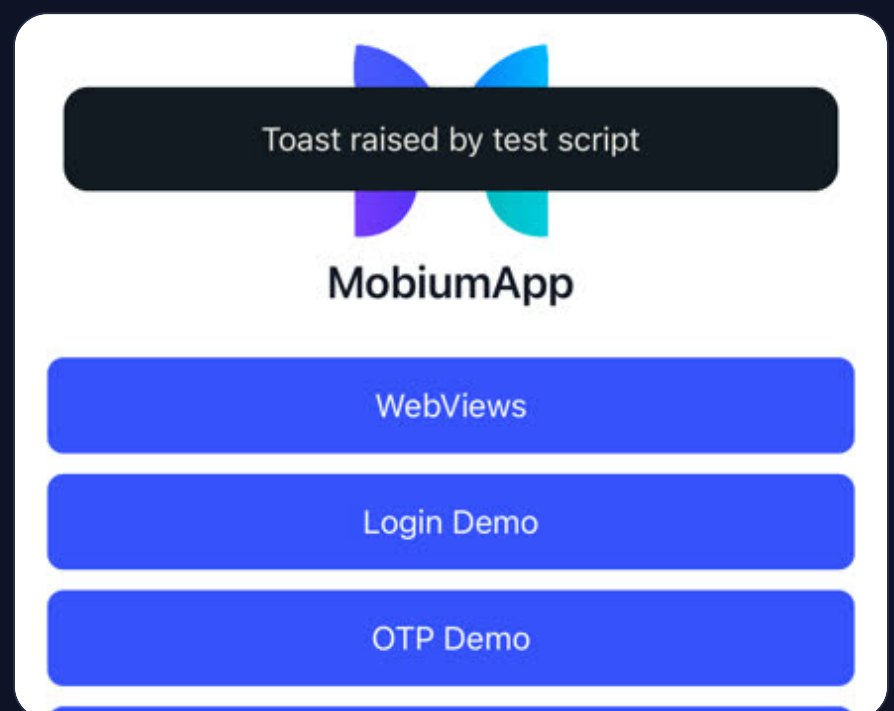
# The test asks the app to do something.

```
// the app, in a build made for testing
GrayBox.register("raiseToast", (message) => {
  ToastAndroid.show(message, ToastAndroid.SHORT);
  return "shown";
});
# the test
device.hook("raiseToast", "Toast raised by test
script") # 'shown'
```

PIXEL 8 PRO



IPHONE 15 PLUS



sign in, seed data, raise a toast



# A door the app opens, and only for the test.

## **The app opts in**

It links a small library and says what it is doing. Work it does not declare is not waited for.

## **Silent unless the test turns it on**

A launch argument on iOS, an intent extra on Android. Anyone opening the same build sees nothing.

## **Only what the app registered**

A hook is called by name. Any other name is refused, with the list of names that exist.

## **Builds made for testing**

Keep hooks out of what ships. An app without the library is driven the ordinary way.



# See the race, then fix it, in a few minutes.

```
$ go install  
github.com/mobiumdev/mobium/cmd/mobium@latest  
$ mobium launch --gray-box dev.mobium.mobiumapp
```

MOBIUM  
[github.com/mobiumdev/mobium](https://github.com/mobiumdev/mobium)

QUICK START, TUTORIAL, EVIDENCE  
[github.com/lana-20/mobium-graybox-demo](https://github.com/lana-20/mobium-graybox-demo)

THE DEMO, WITH EVERY DEVICE'S RECORDINGS  
[lana-20.github.io/mobium-graybox-demo](https://lana-20.github.io/mobium-graybox-demo)